

RAVI KUMAR DUSI

Electronics Engineer

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Langeoog, Germany

in Ravi Dusi ravid-k



WORK EXPERIENCE

Freelancer - AI Domain Specialist -Electronics

Micro1.Inc

Sept 2024 - present

Langeoog

- Personal time from SEP-JAN 2026
- Part-time in Micro1.Inc as AI training expert designing LLM agent with electronics design .
- Apply professional electronics engineering domain expertise to evaluate and optimize next-generation artificial intelligence models
- Generate, review, and refine high-quality technical datasets focusing on electronics circuit design, analog/digital simulations, and PCB layout optimization
- Content creation, Prompt/Rubrics generation to support advanced AI evaluation
- Part-time work in Hotel 'Die Hotel Sandburg' in **Engineering & Maintenance**
- Developed QR-based digital guest information system and exploring LLM tools for workflow automation

Werkstudent Electronics & EMC Engineer

Denpaflux GmbH

Sept 2023 – Sept 2024

Munich

- Designed & simulated PCBs (**Analog/Digital circuits, DC-DC converters, Filters**) using Altium Designer, Implementing EMC rules and layout optimization
- Developed DC-DC buck-boost converters, AC-DC converters, and Sallen-Key filter PCBs, fabricated prototypes, and correlated results between Theory, Simulation, and Lab measurements
- Performed **Signal Integrity / Power Integrity analysis**, controlled impedance routing, and applied IPC/IEC guidelines for high-speed interfaces in my Master's Thesis
- Managed CAD Component Library (symbols, footprints, BOM library) and generated ODB++ and manufacturing documentation for EMC report & Production
- Hands on experience handling electronics lab instruments such as Oscilloscopes, Multimeter, and Vector Network Analyzers, Soldering, Testing

Werkstudent Electronics Engineer

Konux GmbH

Jan 2023 – July 2023

Munich

- Designed a multi-layer test bench system consists of embedded electronics right from conceptualization to production, and assisted in python scripting
- Validated microcontroller-based PCBs for railway monitoring systems, component selection, and circuit dimensioning

SUMMARY

Electronics Engineer with a Master's degree in Electronics , specialized in Microsystems Engineering. Experience in Electronics Design & Development right from conceptualization to production. Fluent in English (C2). Currently seeking a full-time position, willing to relocate, and available to start immediately.

SKILLS

- **Hardware Development:** Analog & mixed-signal circuits, ADC, SI/PI Analysis, Simulation & testing, **EMC fundamentals**.
- **Power Electronics:** DC-DC converters(Buck & Boost), AC-DC Converters, Power distribution, filter design, Grounding, Low Power Design, EMC/ESD compliance.
- **PCB-ECAD:** **Altium Designer**, LibrePCB, KiCAD, DesignSpark, Eagle, **EMC-aware layout**, Multi-layer design, library management , Component selection, PCB Manufacturing.
- **LAB Testing & Validation:** Soldering (THT /SMD), EMC testing, electrical measurement equipment (oscilloscopes, power analyzers, Vector Network Analyzer, DMM, spectrum analyzers, current/voltage sensors)
- **Simulation & Programming:** **LTSpice simulation**, Power & Noise Management, SI/PI analysis, Basic Python & Matlab for data analysis & Lab Automation, knowledge of microcontrollers.
- **Documentation** : Technical documentation, DFM,DFT,Test report, Application notes, MS Office, Github.
- **Failure Analysis & prototyping:** Identification of root causes, Rework, repair of PCB assemblies, test plan preparation, Debugging & Prototyping.
- **Compliance & Standards:** EMC/ESD compliance, IEC 61000-4, IPC 2141A, IPC 2221.

- Maintained CAD library (footprints) to support design-to-manufacture consistency, version control in GitHub/Altium workspace
- Testing of the developed hardware units in the laboratory by Performing soldering, modifying measurement box and cables, optimizing hardware, PCB Assembly, Prototyping, Debugging, supported lab validation and lab automation

Intern Hardware Development

Rocket Factory Augsburg

📅 Sept 2022 – Dec 2022 📍 Augsburg

- Reviewed schematics of Power Distribution Units(PDU)
- Simulation and testing of the developed PDUs in the laboratory
- Assistance in Generating technical documentation & test reports, Also maintaining documentation

EDUCATION

M.Sc Electronics Engineering: MEMS

Hochschule Bremen, Bremen, Germany

📅 (09/2021 – 09/2024)

- Master Thesis : Best practices in Electronics Design. Unveiling myths of common Practices.
- Subjects: Material Science, Micro-technology & Micro-structuring, Analog & Mixed Circuit Design, Concept Engineering of Mixed Technology, Satellite Communication systems.

B.E. Electronics & Communication Engineering

Sathyabama University, Chennai, India

📅 (06/2016 – 03/2020)

- Bachelor Thesis : Design and Development of Unmanned Aerial vehicle for plant disease detection and diagnosis.
- Subjects: Electronic circuits, Electrical network, Control systems, C/C++ programming, Communication systems, Microprocessor, Digital Signal Processing.

LANGUAGES

English

German



Personal skills

Communication skills

Analytical thinking

Problem solving & Presentation skills

Hobbies

PCB designing- personal projects

projects with sensors, BLDC motors

entrepreneurship

Agentic AI development

Archery

Badminton